

# SPATIAL ALIASING-CANCELLATION FOR CIRCULAR MICROPHONE ARRAYS

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## ABSTRACT

Circular microphone arrays facilitate spatial processing and analysis of sound fields in applications where the sound field sources are primarily expected from the azimuthal directions. The operating frequency bandwidth of the array depends on the array aperture and on the number of microphones. At high frequencies, spatial aliasing generates side-lobes in the array beam pattern, which limits array performance. One way to reduce the spatial aliasing effect and to extend the array operating bandwidth is to increase the number of microphones. In this paper, a novel method for designing beamformers that overcome the effect of spatial aliasing is presented. The undesired side-lobes are suppressed through signal processing without increasing the number of microphones or physically modifying the array. This is achieved by taking into account the expected aliasing pattern in the beamformer design, leading to a higher directivity-index at frequencies previously considered to be beyond the microphone array frequency range of operation. A simulation example with a 15-element circular microphone array illustrates the performance of the proposed method.

**Index Terms**— Array processing, Circular microphone arrays, Spatial aliasing, optimal beamformer, aliasing-cancellation, directivity.

## 1. INTRODUCTION

Circular microphone arrays enable two-dimensional spatial recording, signal processing and analysis of sound fields. This is useful in applications such as teleconferencing and outdoor recording where the sound sources are distributed in the horizontal plane [1, 2, 3, 4]. In such applications the circular array may be preferable to volumetric arrays such as the spherical array, as in azimuthal directions it offers better spatial resolution using the same number of microphones. Several circular microphone array configurations have been proposed, including microphones mounted on a rigid object such as a cylinder [3, 4] or a sphere [2], alternatively the microphones can be placed in free field on a virtual ring [5, 6].

Audio recording may require an operating bandwidth that is wider than the array operating frequency range, which is limited by spatial aliasing at high frequencies. Analysis of the performance degradation due to the spatial aliasing error at high frequencies has been previously investigated [4, 7]. Several methods to overcome the spatial aliasing error in circular arrays were proposed, including increasing the number of microphones and placing a microphone at the center of the array [7, 5]. Other methods that were previously suggested for spherical arrays might be applicable for circular arrays and include incorporating microphones with large membranes [8],

or other forms of directional microphones [9, 10]. The shortcoming of all the previously proposed methods is that they involve physical changes to the microphone array.

In this paper a novel method for designing circular array beamformers with reduced performance degradation due to spatial aliasing is presented. The proposed method is based on aliasing-cancellation, previously developed for spherical arrays [11]. It is shown that by taking into account the expected spatial aliasing pattern, a new set of beamforming weights can be designed to reduce the undesired effect of spatial aliasing. The advantage of the proposed method is that aliasing-cancellation is achieved only through signal processing and therefore no physical change in the array is required. This method can therefore be applied to any existing circular array.

This paper is organized as follows. Section 2 reviews the theory of circular microphone array processing. Section 3 describes how high sound field orders are aliased into the lower orders using a new matrix formulation that facilitates the design of the aliasing-cancellation beamformer in section 4. A simulation study is presented in section 5 to illustrate the improved performance achieved by using the new aliasing-cancellation beamformer.

## 2. CIRCULAR ARRAY PROCESSING

The theory of circular array processing is outlined in this section. Consider a ring aperture on the  $xy$ -plane. Using a spherical coordinate system with coordinates  $(r, \theta, \phi)$  [12], the pressure at position  $(R, \frac{\pi}{2}, \phi)$  on a ring with radius  $R$  due to a single unit-amplitude plane wave arriving from direction  $(\theta_o, \phi_o)$  is given by [6]

$$p(k, R, \phi) = e^{jkR \sin \theta_o \cos(\phi_o - \phi)}, \quad (1)$$

where  $j = \sqrt{-1}$  and  $k$  is the wave number. Since functions on the circle are periodic over  $\phi$ , the pressure can be expanded into a set of Fourier series coefficients  $p_m$

$$p_m(k, r) = \frac{1}{2\pi} \int_0^{2\pi} e^{j[kR \sin \theta_o \cos(\phi_o - \phi) - m\phi]} d\phi. \quad (2)$$

The integral can be solved using Bessel's first integral, leading to

$$p_m(k, r) = j^m J_m(kR \sin \theta_o) e^{-jm\phi_o}, \quad (3)$$

where  $J_m(\cdot)$  is the Bessel function of the first kind of order  $m$ . The pressure measured on a ring aperture can now be represented as a Fourier series

$$p(k, R, \phi) = \sum_{m=-\infty}^{\infty} j^m J_m(kR \sin \theta_o) e^{jm(\phi - \phi_o)}. \quad (4)$$

As previously shown [6], at low frequencies, higher orders of the Bessel function  $J_m(kR \sin \theta_o)$  are of very low magnitude and therefore can be neglected. For operating frequencies that satisfy  $kR \leq M$  it can therefore be assumed that the measured pressure is order limited, meaning  $p_m = 0 \forall m > M$ . In this case we can replace the infinite summation in equation (4) with a finite summation over  $m$ ,

$$p(k, R, \phi) = \sum_{m=-M}^M j^m J_m(kR \sin \theta_o) e^{jm(\phi - \phi_o)}. \quad (5)$$

The pressure is now assumed to be sampled at  $Q$  spatial positions using a circular microphone array with microphones located on the ring aperture at  $(R, \phi_q)$ ,  $1 \leq q \leq Q$ . Assuming the pressure coefficients  $p_m$  are limited to order  $M$ , the number of microphones required to reproduce  $p_m$  is  $Q \geq (2M + 1)$ . The pressure coefficients  $p_m$  can be approximated with a summation over spatial samples on the ring aperture [13],

$$\hat{p}_m(k, R) = \sum_{q=1}^Q \alpha_q(k) p(k, R, \phi_q) e^{-jm\phi_q}, \quad (6)$$

where the weighting parameters  $\alpha_q(k)$  are introduced to approximate the Fourier series integral with a summation over the microphones' positions  $(R, \phi_q)$ . The values of  $\alpha_q(k)$  are determined by the sampling scheme. Using the result of equation (6), the array output can be computed as a mode excitation beamformer [1, 6] formulated in the phase mode domain as:

$$y(k) = \sum_{m=-M}^M w_m^* \hat{p}_m(k, R), \quad (7)$$

where the beamformer weights  $w_m$  control the array beam pattern. Different beam patterns can be designed using different sets of beamforming weights. These beamformers work well at frequencies that satisfy  $kR \leq M$ , where the small contribution of higher sound field orders can be neglected. However, at higher frequencies the magnitude of the higher order contributions increases and the performance of these conventional beamformers deteriorates due to spatial aliasing. In the following sections spatial aliasing will be investigated and a new set of optimal beamforming weights will be derived. The new beamformer reduces the effect of spatial aliasing and achieves better performance at high frequencies.

### 3. SPATIAL ALIASING IN CIRCULAR ARRAYS

A circular microphone array of order  $M$  was presented in the previous section. The measured pressure on the circle was assumed to satisfy  $kR \leq M$  and spatial aliasing was therefore neglected. In this paper a method to extend the operating bandwidth is developed. For operating frequencies that satisfy  $M < kR \leq \tilde{M}$ , we assume that the order of the measured pressure coefficients  $p_m(k, R)$  is only limited by  $\tilde{M}$  and that  $p_m = 0 \forall |m| > \tilde{M}$ . The measurement model in Eq. (5) is therefore adapted to support the case where the maximal operating frequency  $kR$  is limited not by the array order  $M$ , but by a higher order  $\tilde{M}$ ,

$$\tilde{p}(k, R, \phi) = \sum_{m=-\tilde{M}}^{\tilde{M}} j^m J_m(kR \sin \theta_o) e^{jm(\phi - \phi_o)}. \quad (8)$$

The same microphone array properties that were described in the previous section are assumed in the following sections; thus a microphone array with  $Q$  microphones of order  $M$  satisfying  $(2M + 1) \leq Q$  is assumed and the number of pressure coefficients  $p_m(k, R)$  that can be reproduced from the measurements is  $2M + 1$ . However, due to the higher frequency, the higher orders of  $p_m(k, R)$  can no longer be neglected and spatial aliasing is introduced in the approximation of  $p_m(k, R)$ . Substituting equation (8) into (6) we get:

$$\begin{aligned} \hat{p}_m(k, R) &= \sum_{m'=-\tilde{M}}^{\tilde{M}} p_{m'}(k, R) \sum_{q=1}^Q \alpha_q e^{-j(m-m')\phi_q} \\ &= \sum_{m'=-\tilde{M}}^{\tilde{M}} p_{m'}(k, R) [\delta_{m-m'} + \varepsilon(m, m')], \end{aligned} \quad (9)$$

where  $\varepsilon(m, m')$  represents the contribution of higher orders to the approximated lower orders and, for the case of a uniform circular array (UCA) where the microphones are equally spaced, with  $\alpha_q = \frac{1}{Q}$  and  $\phi_q = q \frac{2\pi}{Q}$ , we get:

$$\frac{1}{Q} \sum_{q=1}^Q e^{-j(m-m')\frac{2\pi}{Q}q} = \delta_{m-m'} + \varepsilon(m, m'), \quad (10)$$

with the aliasing error elements  $\varepsilon(m, m')$  defined as:

$$\varepsilon(m, m') = \begin{cases} 1 & (m - m') \bmod Q = 0 \text{ and } |m'| > M \\ 0 & \text{else} \end{cases}. \quad (11)$$

For the ideal case, where no aliasing is present, it can be assumed that  $M = \tilde{M}$  and the aliasing error  $\varepsilon(m, m')$  is therefore zero, leading to an exact approximation such that  $\hat{p}_m(k, R) = p_m(k, R)$ . In the general case, however, with  $M < \tilde{M}$ , the aliasing error in the approximation of  $p_m(k, R)$  generates errors in the array output in equation (7). The new measurement model in Eq. (8) facilitates the design of new beamformer weights that suppresses the aliasing error at high frequencies. By taking into account the expected aliasing pattern, the beamformer is designed to suppress the aliasing generated for plane waves arriving from a given angle of elevation. In order to facilitate the computation of the new beamformer weights, equation (8) is rewritten in a matrix form with the new assumptions regarding the higher sound field order:

$$\tilde{\mathbf{p}} = \tilde{\mathbf{B}} \tilde{\mathbf{v}}_o, \quad (12)$$

where  $\tilde{\mathbf{v}}_o = [1, e^{j(\phi_o)}, e^{-j(\phi_o)}, \dots, e^{-j\tilde{M}(\phi_o)}]^T$  is a column vector of length  $(2\tilde{M} + 1)$  that represents a plane wave arriving from direction  $(\pi/2, \phi_o)$ , with the tilde ( $\sim$ ) symbol denoting the higher modes of order  $\tilde{M}$ . The pressure sampled by the microphones is represented by  $\tilde{\mathbf{p}} = [p(k, R, \pi/2, \phi_1), \dots, p(k, R, \pi/2, \phi_Q)]^T$ , a column vector of length  $Q$ , with the elevation angle parameter added, under the assumption that  $\theta = \pi/2$  in this case. Matrix  $\tilde{\mathbf{B}}$  of dimensions  $Q \times (2\tilde{M} + 1)$  is defined in Eq.(13) at the top of the next page.

In order to approximate the pressure coefficient vector  $\tilde{\mathbf{p}}_{\mathbf{m}}$  according to equation (6), the  $(2M + 1) \times Q$  matrix  $\mathbf{C}$  is defined by its elements in row  $m$  and column  $q$  as  $\{\mathbf{C}\}_{m,q} = \alpha_q e^{-jm\phi_q}$ . Now, the pressure coefficient vector  $\hat{\mathbf{p}}_{\mathbf{m}} = [\hat{p}_0, \hat{p}_{-1}, \hat{p}_1, \dots, \hat{p}_M]^T$  can be

$$\tilde{\mathbf{B}} = \begin{bmatrix} J_0(kR) & j^{-1}J_{-1}(kR)e^{-j\phi_1} & j^1J_1(kR)e^{j\phi_1} & \dots & j^M J_M(kR)e^{jM\phi_1} & \dots & j^{\tilde{M}} J_{\tilde{M}}(kR)e^{j\tilde{M}\phi_1} \\ J_0(kR) & j^{-1}J_{-1}(kR)e^{-j\phi_2} & j^1J_1(kR)e^{j\phi_2} & \dots & j^M J_M(kR)e^{jM\phi_2} & \dots & j^{\tilde{M}} J_{\tilde{M}}(kR)e^{j\tilde{M}\phi_2} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ J_0(kR) & j^{-1}J_{-1}(kR)e^{-j\phi_Q} & j^1J_1(kR)e^{j\phi_Q} & \dots & j^M J_M(kR)e^{jM\phi_Q} & \dots & j^{\tilde{M}} J_{\tilde{M}}(kR)e^{j\tilde{M}\phi_Q} \end{bmatrix}. \quad (13)$$

computed, as shown in equation (6), by multiplying equation (12) by matrix  $\mathbf{C}$

$$\hat{\mathbf{p}}_{\mathbf{m}} = \mathbf{C}\tilde{\mathbf{B}}\tilde{\mathbf{v}}_o. \quad (14)$$

In the ideal case, where no aliasing is present, such that  $M = \tilde{M}$  in equation (9), we expect that  $\hat{\mathbf{p}}_{\mathbf{m}} = \mathbf{p}_{\mathbf{m}}$ . For a circular array with an irregular sampling scheme, matrix  $\mathbf{C}$  can be calculated using the matrix formulation in equation (14), as previously shown [13, 11]. In the following sections a UCA with  $\alpha_q = \frac{1}{Q}$  is assumed, thus the elements of matrix  $\mathbf{C}$  are as shown above. The result of the matrix multiplication  $\mathbf{C}\tilde{\mathbf{B}}$  is a  $(2M+1) \times (2\tilde{M}+1)$  matrix that holds the information about the spatial aliasing pattern. The left  $(2M+1)$  columns reduce to a diagonal matrix which describes the relation between the arriving sound-field mode and the corresponding pressure coefficient  $p_m$ . However, the right side of the matrix represents the contribution of sound-field modes of higher orders that are aliased into the lower orders, as in equation (9). The aliasing error elements are structured to form the aliasing pattern previously described [7, 9]. Based on this aliasing pattern, a new beamformer design will be presented in the next section.

#### 4. BEAMFORMER DESIGN WITH OPTIMAL ALIASING-CANCELLATION

The array output in equation (7) is now presented in a matrix form and, using (14), can be rewritten as:

$$y(k) = \mathbf{w}_{\mathbf{m}}^H \mathbf{C}\tilde{\mathbf{B}}\tilde{\mathbf{v}}_o, \quad (15)$$

where  $\mathbf{w}_{\mathbf{m}} = [w_0, w_{-1}, w_1, \dots, w_M]^T$  is a column vector containing the beamforming weights  $w_m$  in equation (7). The aliasing error in the approximation of the pressure coefficient vector  $\mathbf{p}_{\mathbf{m}}$  in equation (14) results in the formation of grating-lobes in the array beam pattern. In order to reduce this aliasing error, a new set of beamformer weights  $\mathbf{w}_{\mathbf{m}}$  are calculated to maximize the directivity of the array. The main difference between the conventional maximum directivity beamformer [14] and the new aliasing-cancellation beamformer is that the aliasing error is taken into account in the computation of the beamformer weight vector  $\mathbf{w}_{\mathbf{m}}$ . In the next sections it will be shown that this choice of  $\mathbf{w}_{\mathbf{m}}$  reduces the effect of aliasing and suppresses the grating-lobes. In order to calculate the directivity, a sound field composed of a single unit-amplitude plane wave arriving from direction  $(\pi/2, \phi_o)$  with modes of orders up to  $\tilde{M}$  is assumed. The directivity factor  $Q$  is evaluated in the  $xy$ -plane to support the new spatial aliasing model and can be written as:

$$Q = \frac{|y(k, \pi/2, \phi_l)|^2}{\frac{1}{2\pi} \int_0^{2\pi} |y(k, \pi/2, \phi_o)|^2 d\phi_o}, \quad (16)$$

where the numerator represents the array output in which the incident plane wave arrival direction equals the array look direction  $(\theta_l, \phi_l)$ . Using equation (15) with  $\tilde{\mathbf{v}}_o = \tilde{\mathbf{v}}_l$  to indicate the equality  $(\theta_o, \phi_o) = (\theta_l, \phi_l)$ , the expression for the numerator of equation (16) can be written as:

$$|y(k, \pi/2, \phi_l)|^2 = \mathbf{w}_{\mathbf{m}}^H \mathbf{C}\tilde{\mathbf{B}}\tilde{\mathbf{v}}_l \tilde{\mathbf{v}}_l^H \tilde{\mathbf{B}}^H \mathbf{C}^H \mathbf{w}_{\mathbf{m}}. \quad (17)$$

The denominator in equation (16) represents the array output due to isotropic noise, which refers in this case to plane waves arriving from all azimuthal directions with equal power density. Using the expression for the array output in equation (15), the denominator can be written as:

$$\begin{aligned} & \frac{1}{2\pi} \int_0^{2\pi} |y(k, \pi/2, \phi_o)|^2 d\phi_o \\ &= \frac{1}{2\pi} \mathbf{w}_{\mathbf{m}}^H \mathbf{C}\tilde{\mathbf{B}} \int_0^{2\pi} \tilde{\mathbf{v}}_o \tilde{\mathbf{v}}_o^H d\phi_o \tilde{\mathbf{B}}^H \mathbf{C}^H \mathbf{w}_{\mathbf{m}} \\ &= \mathbf{w}_{\mathbf{m}}^H \mathbf{C}\tilde{\mathbf{B}}\tilde{\mathbf{B}}^H \mathbf{C}^H \mathbf{w}_{\mathbf{m}}. \end{aligned} \quad (18)$$

Substituting equations (17) and (18) into (16) we get:

$$Q = \frac{\mathbf{w}_{\mathbf{m}}^H \mathbf{C}\tilde{\mathbf{B}}\tilde{\mathbf{v}}_l \tilde{\mathbf{v}}_l^H \tilde{\mathbf{B}}^H \mathbf{C}^H \mathbf{w}_{\mathbf{m}}}{\mathbf{w}_{\mathbf{m}}^H \mathbf{C}\tilde{\mathbf{B}}\tilde{\mathbf{B}}^H \mathbf{C}^H \mathbf{w}_{\mathbf{m}}}. \quad (19)$$

Next, we wish to find the beamformer weight vector  $\mathbf{w}_{\mathbf{m}}$  that maximizes the directivity factor  $Q$ . Equation (19) is rewritten in a generalized Rayleigh quotient form [14], by substituting  $\mathbf{G} = \mathbf{C}\tilde{\mathbf{B}}\tilde{\mathbf{v}}_l \tilde{\mathbf{v}}_l^H \tilde{\mathbf{B}}^H \mathbf{C}^H$  and  $\mathbf{H} = \mathbf{C}\tilde{\mathbf{B}}\tilde{\mathbf{B}}^H \mathbf{C}^H$ :

$$Q = \frac{\mathbf{w}_{\mathbf{m}}^H \mathbf{G} \mathbf{w}_{\mathbf{m}}}{\mathbf{w}_{\mathbf{m}}^H \mathbf{H} \mathbf{w}_{\mathbf{m}}}. \quad (20)$$

As previously presented [14], the maximum value of the generalized Rayleigh quotient is the largest generalized eigenvalue of the equivalent generalized eigenvalue problem:

$$\mathbf{G} \mathbf{v}_{\lambda} = \lambda \mathbf{H} \mathbf{v}_{\lambda}, \quad (21)$$

where  $\lambda$  is the generalized eigenvalue and  $\mathbf{v}_{\lambda}$  is the corresponding generalized eigenvector. The largest eigenvalue and the corresponding eigenvector will hold the value of the maximal directivity factor  $Q_{max} = \lambda_{max}$  and the beamformer weight vector  $\mathbf{w}_{\mathbf{m}} = \mathbf{v}_{\lambda_{max}}$  that maximizes the directivity. Since matrix  $\mathbf{G} = \mathbf{g} \mathbf{g}^H$  is composed of the  $(2M+1) \times 1$  vector  $\mathbf{g} = \mathbf{C}\tilde{\mathbf{B}}\tilde{\mathbf{v}}_l$ ,  $\mathbf{G}$  has only one eigenvector and one eigenvalue. The maximum directivity for the optimal beam pattern is therefore

$$Q = \mathbf{g}^H \mathbf{H}^{-1} \mathbf{g} = \tilde{\mathbf{v}}_l^H \tilde{\mathbf{B}}^H \mathbf{C}^H (\mathbf{C}\tilde{\mathbf{B}}\tilde{\mathbf{B}}^H \mathbf{C}^H)^{-1} \mathbf{C}\tilde{\mathbf{B}}\tilde{\mathbf{v}}_l \quad (22)$$

and the beamforming weight vector  $\mathbf{w}_{\mathbf{m}}$  that maximizes the directivity is given by:

$$\mathbf{w}_{\mathbf{m}} = \arg \max_{\mathbf{w}_{\mathbf{m}}} Q = \mathbf{H}^{-1} \mathbf{g} = (\mathbf{C}\tilde{\mathbf{B}}\tilde{\mathbf{B}}^H \mathbf{C}^H)^{-1} \mathbf{C}\tilde{\mathbf{B}}\tilde{\mathbf{v}}_l. \quad (23)$$

Another measure to assess the array performance is the white-noise gain (WNG), which evaluates the improvement in SNR at the array output compared with the array input. In some cases [15], increasing the directivity comes at the expense of decreasing the WNG. We shall therefore verify the WNG produced by the new aliasing-cancellation beamformer. The WNG is given by [6, 13]

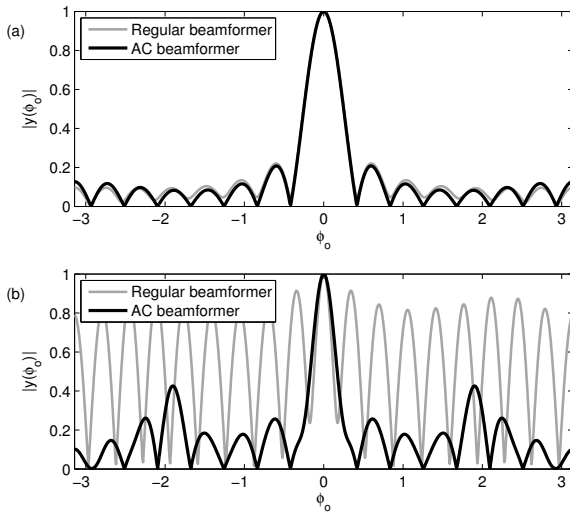
$$A_w = \frac{|y(k, \theta_l, \phi_l)|^2}{\mathbf{w}_{\mathbf{m}}^H \mathbf{w}_{\mathbf{m}}}, \quad (24)$$

where  $\mathbf{w}$  is the beamforming weights vector in the space-domain defined as  $\mathbf{w}^H = \mathbf{w}_m^H \mathbf{C}$ , see equation (15).

In the next section the aliasing-cancellation beamformer is compared to a conventional maximum directivity beamformer [14], referred to as a regular beamformer. The regular beamformer weights can be calculated using equation (23) with  $M = \bar{M}$ , such that only the  $(2M + 1)$  left columns of matrix  $\tilde{\mathbf{B}}$  are used instead of the complete  $\tilde{\mathbf{B}}$  matrix.

## 5. DESIGN EXAMPLE

A design example is presented in this section to evaluate the performance of the proposed method. Consider a circular microphone array of radius  $R = 10$  cm in free field with  $Q = 15$  microphones distributed uniformly on the circle, providing an array order  $M = 7$ . In order to satisfy the aliasing-free condition, the usable operating frequency of the array is limited by  $kR \leq M = 7$ , leading to  $f_{max} = 3.8$  KHz. The optimal aliasing-cancellation beamforming weight vector  $\mathbf{w}_m$  is calculated according to Eq. (23). First, the regular beamformer and the optimal aliasing-cancellation beamformer are compared at two operating frequencies. Second, the directivity factor  $Q$  and the WNG of these two beamformers is compared over a wider range of frequencies.

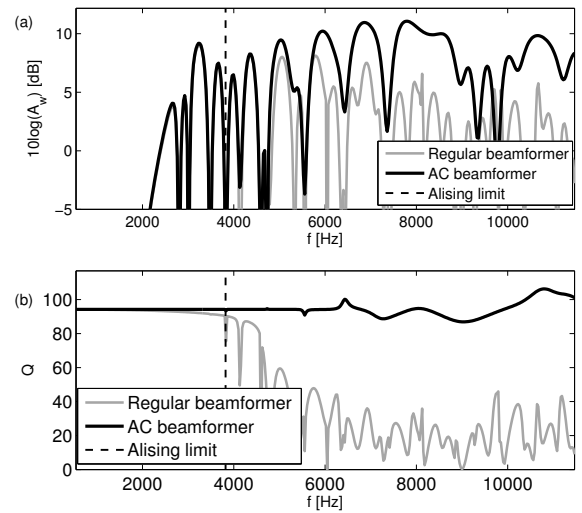


**Fig. 1.** Magnitude of seventh order array beam pattern; comparison between optimal and regular beamformers at (a)  $f = 3230$  Hz ( $kR = 6$ ) and (b)  $f = 5460$  Hz ( $kR = 10$ )

Figure 1 shows a comparison between the regular beam pattern and the aliasing-cancellation beam pattern with look direction  $(\theta_l, \phi_l) = (\pi/2, 0)$  at a relatively low frequency where the spatial aliasing can be neglected. The beam pattern of the two beamformers is practically identical, as shown in Figure 1(a). However, at a higher frequency, where the spatial aliasing is significant, the regular beam pattern is not usable, while the aliasing-cancellation beam pattern preserves high performance as shown in Figure 1(b).

A comparison between the two beamformers is further pursued over a wider bandwidth. The WNG and the directivity factor  $Q$  of the two beamformers with the same look direction as in the first example are presented in Fig.2(a) and (b), respectively. The WNG of the two beamformers is compared in Figure 2(a), which illustrates that the WNG increases when using the aliasing-cancellation

beamformer. The regular beamformer and the aliasing-cancellation beamformer are calculated to maximize the directivity and therefore the WNG of these beamformers is not expected to be optimal. Furthermore, the microphones are placed in free field and the WNG is therefore very low at frequencies related to the zeros of the Bessel functions. Avoiding the low WNG around those frequencies can be achieved by placing the microphones around a rigid object [3, 4, 2]. The directivity factor is presented in Figure 2(b). The figure shows that as the frequency increases and spatial aliasing becomes more significant, the directivity of the regular beamformer decreases while the directivity of the aliasing-cancellation beamformer remains high. This result verifies the advantage of the proposed beamformer over a wide frequency bandwidth. The increase in the WNG, together with the increase in directivity, implies that the aliasing-cancellation beamformer achieves an improvement in performance and displays robustness to noise.



**Fig. 2.** Comparison between beamformers (a) array WNG, (b) array directivity factor  $Q$

## 6. CONCLUSION

A matrix formulation for describing a high-order sound field aliased into the lower array orders was derived. This formulation was found to be useful for the design of an optimal aliasing-cancellation beamformer, which achieves higher directivity compared with a conventional beamformer. The improvement in performance is achieved at high frequencies which were previously considered out of the microphone array operating range, therefore increasing the circular array bandwidth. The proposed design does not require physical changes in the array and could therefore be applied within any existing circular array. Results of a simulation that demonstrate the improved performance of the proposed beamformer were presented. Conducting an experimental study of the optimal aliasing-cancellation beamformer with different types of circular array is proposed for future work.

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